

Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 8, 2001

Instructions: Answer all four questions. Open book and open notes.

1. At faculty meetings we get pizza. The types are plain, veggie, and pepperoni. The probability of getting veggie is twice of that of getting plain. The probability of getting pepperoni is the same as plain.

(a) If three pizzas are ordered at random, what is the probability of getting one of each kind?

(b) If four are ordered, what is the probability that we get at least one of each kind?

(c) If four are ordered, what is the probability that they are all plain?

2. Let X_1, X_2, X_3 be independent binomial variables with parameters $(1, \frac{1}{2}), (2, \frac{1}{2}), (3, \frac{1}{3})$, respectively.

(a) Find $P\{X_1 = 1, X_2 = 1, X_3 = 1\}$.

(b) Find the covariance of $Y_1 = X_1 + X_2$ and $Y_2 = X_2 + X_3$.

3. (a) We seek the difference in probabilities of hitting a home run for Mark McGuire and Sammy Sossa. The difference is $p_m - p_s$. Based on data for 1998, Mark hit 70 homers in 600 at bat while Sammy hit 65 (I think) in 690 at bat. Find a 99% confidence interval for the difference.

(b) Let X_1, X_2, X_3 be independent $N(\mu, \sigma^2)$, μ and σ^2 unknown. Find a 98% confidence interval for σ^2 given $X_1 = 1, X_2 = 0, X_3 = -1$.

4. Let X be a single observation from a population whose density is

$$f_X(x; \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the most powerful test of size $\alpha = .05$ for $H_0 : \theta = 1$ vs $H_1 : \theta = 2$.